

Learning to act in high-dimensional spaces

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Keywords:

Summary

Summary. TBD.

Contribution(s)

1. Provide a succinct but precise list of the contribution(s) of the paper. Use contextual notes to avoid implications of contributions more significant than intended and to clarify and situate the contribution relative to prior work (see the examples below). If there is no additional context, enter “None”. Try to keep each contribution to a single sentence, although multiple sentences are allowed when necessary. If using complete sentences, include punctuation. If using a single sentence fragment, you may omit the concluding period. A single contribution can be sufficient, and there is no limit on the number of contributions. Submissions will be judged mostly on the contributions claimed on their cover pages and the evidence provided to support them. Major contributions should not be claimed in the main text if they do not appear on the cover page. Overclaiming can lead to a submission being rejected, so it is important to have well-scoped contribution statements on the cover page.

Context: None

2. The submission template for submissions to RLJ/RLC 2026

Context: Built from previous RLC/RLJ, ICLR, and TMLR submission templates

3. *[Example of one contribution and corresponding contextual note for the paper “Policy gradient methods for reinforcement learning with function approximation” (Sutton et al., 2000).]*

This paper presents an expression for the policy gradient when using function approximation to represent the action-value function.

Context: Prior work established expressions for the policy gradient without function approximation (Williams, 1992).

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Abstract

Reinforcement learning (RL) performs strongly on continuous-control benchmarks, yet many real-world systems such as articulated robots, musculoskeletal models, and dexterous hands require controlling tens to hundreds of actuators. In this regime, action dimensionality can be a dominant bottleneck, but existing benchmarks often entangle dimensionality with changes in objectives, morphology, or contact dynamics, obscuring how algorithms truly scale. We introduce HiGym, a high-dimensional continuous-control benchmark built on DMControl. HiGym comprises five environment families with multiple tasks and five embodiment complexities. We isolate action-space scaling by keeping tasks and dynamics as similar as possible while increasing degrees of freedom via a principled embodiment-scaling procedure that recursively splits joints to reach target dimensionalities. This yields matched task variants that differ primarily in action dimension. Across PPO, FastTD3, FastSAC, TD-MPC2, and TD-M(PC)², we find that model-based methods degrade substantially less as dimensionality increases, preserving higher sample-efficiency and asymptotic performance than model-free baselines.

1 Introduction

2 Related Works

3 HiGym: A continuous control benchmark for DoF scaling

We present HiGym, a benchmark for studying how reinforcement learning algorithms scale with action dimensionality. HiGym is built on top of the DeepMind Control Suite and consists of six environment families spanning diverse morphologies: legged locomotion (Quadruped, Cheetah, Walker), whole-body humanoid control (Humanoid CMU), and planar manipulation (Finger, Reacher). Each family of embodiments is obtained by a principled *joint-splitting* procedure that progressively subdivides existing joints to reach target dimensionalities. Crucially, the task objective, reward function, and overall morphological structure remain fixed across sizes within a family; only the number of actuated degrees of freedom changes. This design isolates the effect of action dimensionality from confounding factors such as changes in contact dynamics, observation semantics, or reward shaping that would arise from switching between unrelated environments.

The joint-splitting procedure operates as follows: Given a base model with n actuated joints per kinematic chain, we subdivide each chain segment into finer sub-segments by inserting intermediate bodies at equal arc-length positions along the original limb geometry. Doing this leads to the following further modifications to isolate scaling effects and preserve physical properties:

- Each new joint inherits the axis conventions of its parent, and the total curvature budget (sum of joint ranges) and torque budget (sum of gear ratios) per chain are preserved by scaling individual ranges and gears by $1/k$ when a single joint is split into k sub-joints.
- Geom masses are redistributed so that the total limb mass is exactly conserved.

Table 1: **HiGym benchmark suite.** Six environment families with their base DoF, scaled DoF variants, and associated tasks.

Environment	Base DoF	Scaled DoFs	Tasks
Quadruped	16	16, 32, 64, 128, 256	walk, run, escape, fetch
Cheetah	6	4, 8, 16, 32, 64, 128, 256	run
Walker	6	4, 8, 16, 32, 64, 128, 256	stand, walk, run
Finger	2	2, 4, 8, 16, 32, 64	spin, turn_easy, turn_hard
Reacher	2	2, 4, 8, 16, 32, 64	easy, hard
Humanoid CMU	56	80	stand, walk, run



Figure 1: **Quadruped embodiment scaling.** The four-legged morphology at 16 to 256 DoF. Joint density increases while overall limb geometry is preserved.

• Collision filtering is applied to intermediate bodies to prevent spurious self-contacts without shrinking segment radii (see appendix for more details).

This yields a family of matched environments in which a d -DoF variant and a $2d$ -DoF variant share the same total reach, the same peak torque per limb, and the same total body mass. With these changes, we were able to successfully scale up quadruped, cheetah and walker embodiments to 256 DoF (compared to default 6), and also present a competing frontier in whole-body humanoid control by scaling up the arms and legs of CMU humanoid to a total of 80 DoF from 56. Table 1 summarises the six environment families, their DoF ranges, and the tasks associated with each. We will now describe each embodiment below.

3.1 Embodiments and tasks

Quadruped. The quadruped is a four-legged torso-driven locomotor based on the DMControl quadruped model. The base variant has 16 DoF (4 joints per leg: 1 yaw + 3 pitch). Scaling proceeds by subdividing each leg into more segments along the original hip-to-toe arc, yielding variants at 32, 64, 128, and 256 DoF. Pitch joint ranges are scaled by $(3/n_{\text{pitch}})$ to preserve the total curvature budget per leg, and actuator gains follow a compensated power law $1000 \cdot (4/n_{\text{joints}})^{0.4}$ to maintain consistent peak torque. Leg mass is held constant across sizes by distributing the original limb mass equally among segments. Only toe geoms participate in ground contact; intermediate capsules are collision-filtered. Tasks include `walk` and `run` (reward forward velocity while maintaining upright posture and a target torso height), `escape` (navigate a hilly terrain), and `fetch` (retrieve a ball).

Cheetah. The cheetah is a planar biped (sagittal-plane runner) with two legs attached to a rigid torso. The base model has 6 DoF (3 joints per leg). Each leg is subdivided along front and back leg arcs to produce variants at 4, 8, 16, 32, 64, 128, and 256 DoF. Joint ranges scale as $(3/n)^y$ and gear ratios as $(3/n)^x$, where n is the number of joints per leg. To compensate for the forward centre-of-mass shift induced by the head, the back-leg mass is scaled by a factor of $1.3\times$ and total body mass is normalised to 14 kg via MuJoCo’s `settotalmass`. Only a single task is defined: `run`, which rewards forward speed with a linear tolerance.

Walker. The walker is a planar biped that must balance and locomote on two symmetric legs. The base model has 6 DoF (3 joints per leg: hip, knee, ankle). Scaling subdivides each leg along the hip-to-foot arc, yielding variants at 4, 8, 16, 32, 64, 128, and 256 DoF. Joint ranges are scaled by $(3/n)$

66 to preserve total curvature per leg, while gear ratios remain unscaled. Reward is a composite of torso
67 height, upright orientation, and forward velocity (for `walk` and `run`) or standing stability alone (for
68 `stand`). Episode initialisation places the walker in a near-upright pose with small joint-angle noise.

69 **Finger.** The finger environment features a planar serial-chain manipulator that must interact with
70 a freely rotating spinner body. The base model has 2 DoF (proximal and distal hinge joints). Scaling
71 extends the chain to 4, 8, 16, 32, and 64 DoF by subdividing the finger into finer segments along
72 its arc. Gravity is disabled, so the task is purely about contact-mediated manipulation. Gear ratios
73 are interpolated along the chain (30 at the base, 15 at the tip) and scaled by $(2/n)^{0.55}$ to preserve
74 the torque-to-inertia ratio; for the largest variant ($n=64$), additional joint stiffness is introduced to
75 prevent buckling. The spinner’s inertia is also scaled with DoF’s gear scaling to maintain a consistent
76 difficulty gradient. Three tasks are defined: `spin` (achieve a target angular velocity of the spinner),
77 `turn_easy`, and `turn_hard` (rotate the spinner to a target angle with varying tolerances).

78 **Reacher.** The reacher is a planar serial-chain arm that must position its fingertip on a randomised
79 target. The base model has 2 DoF (shoulder and wrist hinges). Scaling extends the chain to 4, 8,
80 16, 32, and 64 DoF while holding the total arm length fixed at 0.24 m. The shoulder joint is left
81 unlimited (free rotation for aiming) while non-shoulder joints distribute the total curvature budget
82 of 160° equally. Restoring stiffness and damping are tuned per variant to prevent the chain from
83 collapsing under its own compliance. Two tasks are defined: `easy` and `hard`, which differ in the
84 target radius.

85 **Humanoid CMU.** The humanoid CMU is the most complex embodiment in HiGym, based on the
86 56-DoF CMU motion-capture skeleton from DMControl. Unlike the other families, which scale by
87 subdividing existing chains, the humanoid is extended to 80 DoF by inserting 24 intermediate bodies
88 at the midpoints of the major limb segments (thigh, shin, foot, upper arm, forearm, and shoulder,
89 bilaterally). Each intermediate joint inherits the rotation axes of its parent, and the original joint
90 ranges, gear ratios, and geom masses are split 50/50 between parent and child to exactly preserve
91 the total curvature budget, torque budget, and body mass. Stiffness and damping classes are inherited
92 from the parent joint. Three tasks are defined: `stand`, `walk`, and `run`.

93 A The first appendix

94 This is an example of an appendix.

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96 subject to the 8–12 page limit, are peer reviewed, and can contain content central to the claims of
97 the paper.

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103 Acknowledgments

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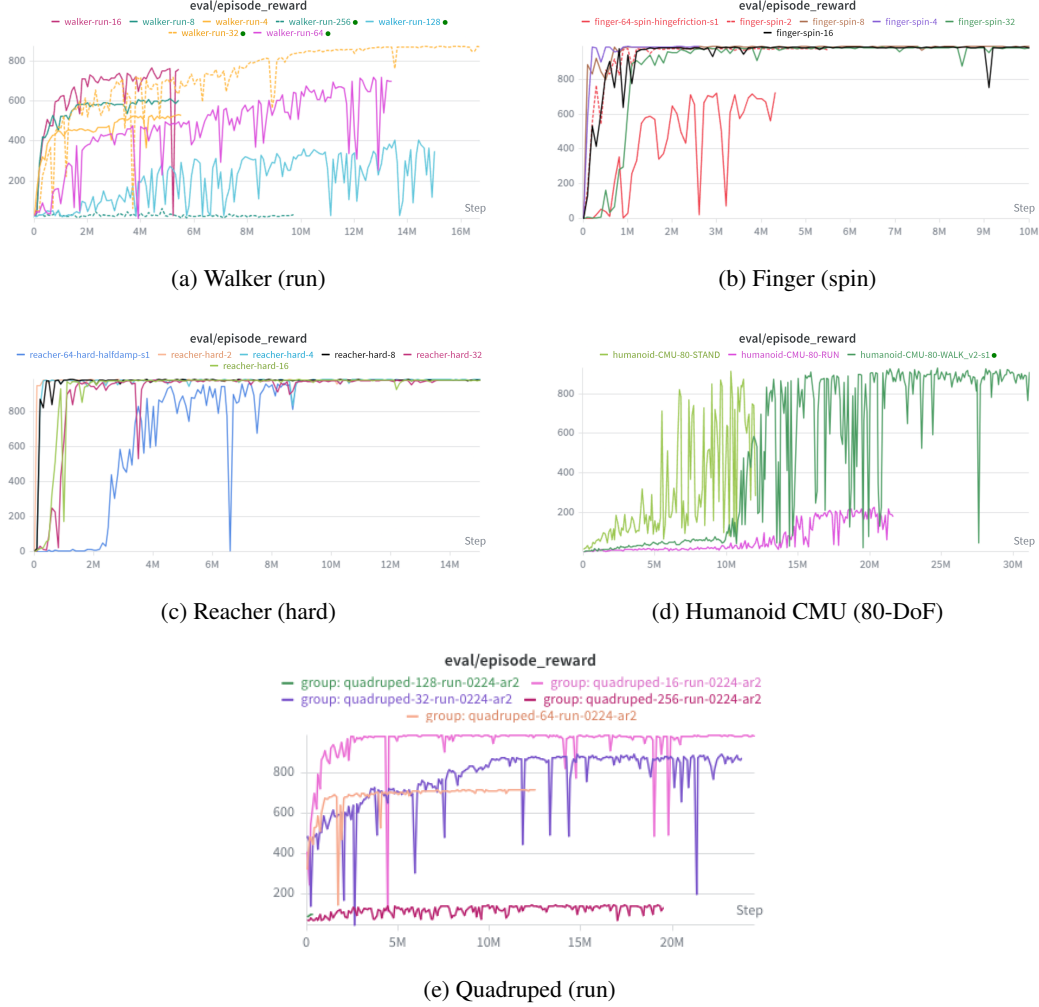


Figure 2: **Embodiment scaling results.** Episode reward as a function of training steps across DoF variants for four environment families. Higher DoF generally increases sample complexity, though the rate of degradation varies across morphologies.

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Supplementary Materials

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